Nile University

School of Engineering & Applied Sciences

PHY 111 - Physics I
Fall 2026

Course Description:

This is the first course in a two-course series that provides the university physics foundation for engineering students. This course gives a solid theoretical and practical knowledge of Material properties and electromagnetics.

Instructors

Dr. Mai Sallam

Dr. Nouran Ali

Dr. Essam Gomaa

Prof. Yasser El-Batawy

Textbooks

- **J. Jewett and R. Serway**, "*Physics for Scientists and Engineers,*" *Thomson, 9th edition*
- **Giancoli**, "*Physics: Principles with Applications,*" *Global Edition, 7th edition*

Course Outline:

Part I: Material Properties

- Dimensions and Units
- Elasticity
- Fluid Mechanics

Part II: Oscillatory Motion and Waves

- **Oscillatory Motion:** Simple Harmonic Motion, Motion of an object attached to a spring, mathematical representation of a simple harmonic motion, energy of the simple harmonic oscillator, comparing simple harmonic motion with uniform circular motion.

- **Wave Motion:** Propagation of a disturbance, sinusoidal waves, the speed of waves on a string, the linear wave equation, harmonic traveling wave, wave intensity, superposition of waves, sound wave, Doppler effect.

Part III: Thermodynamics

- **Temperature:** Temperature, thermometers, absolute temperature scale, thermal expansion of solids and liquids.
- **Heat and the First Law of Thermodynamics:** Heat and internal energy, specific heat, latent heat, work and heat in thermodynamic processes, the first law of thermodynamics
- **Heat Engines and The Second Law of Thermodynamics:** Heat engines and the second law of thermodynamics, heat pumps and refrigerators, reversible and irreversible processes, Carnot engine, thermal cycles.

Grading System

Class Attendance & Participation	**5 %**
Quizzes	**10 %**
Assignments	**10 %**
Midterm Exam 1	**15 %**
Midterm Exam 2	**15 %**
Lab sessions	**5 %**
Lab Final Exam	**8 %**
Final Exam	**32 %**

Policy on Digital Devices

The use of digital devices such as mobile phones, tables, and laptops is prohibited during the class time, unless the instructor authorizes their use in connection with the course material. If you use a laptop or a tablet for note taking, you are requested to sit in the front raw.

Academic Integrity

1. Statement Regarding Academic Honesty Nile University

requires that all students' work is performed honestly and without unauthorized assistance. Undergraduate and graduate students, by registration, acknowledge the jurisdiction of University policies regarding academic dishonesty as documented in the Student Handbook. These policies and procedures to all full- and part-time students enrolled in undergraduate and graduate courses on-campus, off-campus, and via distance learning. The details of the University's policies can be found on the university website. A failing grade received for reasons of academic dishonesty will not be subject to repeat approval, and a student who violates rules for examination dishonesty may be required to withdraw from the University.

2. Statements for Academic Accommodations for Students with Disabilities

Any student with a disability who needs a classroom accommodation, access to technology or other academic assistance in this course should contact the instructor.

3. Statement Defining Expectations for Classroom Conduct

All student activities in the University, including this course, are governed by the University Policies and Procedures as defined in the Student Handbook. Students who engage in behavior that disrupts the learning environment may be asked to leave the classroom.

4. Cheating

"Cheating means getting unauthorized help on an assignment, quiz or examination."

- You must not receive from any other students or give to any other students any information, answers, or help during an exam.
- You must not use unauthorized sources for answers during an exam. You must not take notes or books to the exam when such aids are forbidden, and you must not refer to any book or notes while you are taking the exam unless the instructor indicates it is an "open book" exam.
- You must not obtain exam questions illegally before an exam or tamper with an exam after it has been corrected.

5. Plagiarism

"Plagiarism" means submitting work as your own that is someone else's. For example, copying material from a book, the Internet, or another source without acknowledging that the words or ideas are someone else's and not your own is plagiarism. If you copy an author's words exactly, treat the passage as a direct quotation and supply the appropriate citation. If you use someone else's ideas, even if you paraphrase the wording, appropriate credit should be given. You have committed plagiarism if you purchase a term paper or submit a paper as your own that you did not write.